
When Does Action-Type Supervision Help GUI Grounding?

Separating Real Effects from Evaluation Artifacts

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Abstract

1 GUI agents must predict an action type (*click*, *scroll*, *type*, ...) and a spatial
2 target from a screenshot and an instruction. We study whether explicitly factor-
3 ing these decisions helps a 2B-parameter agent, via a matched-compute compar-
4 ison of five conditioning mechanisms on Qwen2-VL: a flat baseline (A), an auxil-
5 iary action-type loss (B), hard routing (C), an additive embedding hook (D-hook),
6 and the prepended learned action embedding our original proposal advocated (D-
7 token). Our first evaluation of this question was wrong in three independent ways,
8 each of which we diagnose, quantify, and repair: (i) injecting conditioning via
9 `inputs_embeds` silently bypasses Qwen2-VL’s M-RoPE position path and man-
10 ufactured an 8σ *false negative*; (ii) train/validation splits taken as consecutive
11 slices of one episode stream leaked shared task goals (24% of validation steps
12 at the headline setting, up to 85% at scale) while looking clean at the episode
13 level (2%); and (iii) a paired bootstrap pooling (seed, example) predictions as in-
14 dependent units was anti-conservative, rejecting a true clustered null 21.2% of the
15 time at nominal 5% (an episode-clustered replacement achieves 5.1%). We then
16 rebuilt the data pipeline around frozen goal-disjoint splits with a test region no
17 phase of the project had ever touched, prespecified four contrasts, and evaluated
18 once. **Every mechanistic conclusion reversed.** The auxiliary loss that the re-
19 paired statistics still endorsed on historical data (B–A +0.051 [+0.020, +0.081],
20 $p = 0.0043$, versus the invalid pooled $p = 0.0002$) is null on the untouched test
21 set (+0.011 [-0.005, +0.029], Holm $p = 0.309$); the prepended action embedding
22 our proposal advocated and the course-era version reported as *refuted* in fact beats
23 it (D-token–B +0.035 [+0.017, +0.054], Holm $p = 0.0027$); the gain is propor-
24 tionally largest on *scroll* ($2.5\times$) rather than *click*; the control condition designed to
25 show no effect shows one, and is therefore invalid; and the deployable predicted-
26 type pipeline is null (+0.015 [-0.004, +0.034], Holm $p = 0.309$) because Stage 1
27 accuracy falls from 0.795 on validation to 0.708 on unseen goals, consuming
28 the entire oracle gain. What survives is narrow and honest: *oracle* action-type
29 conditioning improves grounding (D-hook–A +0.057 [+0.035, +0.079], Holm
30 $p = 0.0004$), and nothing we can actually deploy does. We release the frozen
31 splits, per-example predictions, and the clustered-inference code. A study can be
32 multi-seed, ablated, controlled, and significant at $p < 0.002$ and still have all of
33 its mechanistic claims invert under a clean split and a calibrated test.

1 Introduction

A GUI agent maps a screenshot and a natural-language instruction to a low-level action: a type (*click*, *scroll*, *type*, *drag*, ...) and, for spatial actions, a pixel-level target. State-of-the-art native agents (UI-TARS [13], OS-Atlas [11], SeeClick [4], ShowUI [7]) emit both in one autoregressive stream, entangling the two decisions. A documented failure mode follows: commit to the wrong action type, then ground *coherently* within the wrong semantic frame – scrolling past a visible target instead of clicking it. This motivated our original hypothesis, formulated as an architecture bet: a learned action-type embedding, prepended to the instruction, should let a grounding model condition on *what* before deciding *where*, and should help most in low data.

This paper reports what happened when we tested that bet carefully, and it has two intertwined subjects. The first is the scientific question: *when does action-type supervision change GUI grounding, and through which mechanism?* The second is methodological: *which implementation and evaluation choices can create a false appearance of benefit or harm in studies of exactly this kind?* We came to the second subject involuntarily – three separate artifacts in our own pipeline each produced a wrong conclusion at some point during the project – and we believe the post-mortem is at least as useful to practitioners as the object-level result.

Concretely, we compare five matched-compute Stage-2 conditioning mechanisms on Qwen2-VL-2B [10] with LoRA, over AITW [9] with a Mind2Web [5] cross-dataset control, preceded by a Stage-1 action-type classifier study. Our contributions:

- **A confirmatory answer, and it is mostly negative.** On an untouched, goal-disjoint test set: auxiliary action-type supervision does not measurably help; *oracle* inference-time conditioning does (+0.057 [+0.035, +0.079]); the deployable predicted-type pipeline does not, because the action-type classifier loses 0.795→0.708 accuracy on unseen task goals; and the proportionally largest effect is on *scroll*, not the click disambiguation our exploratory analysis reported.
- **Three quantified evaluation pitfalls.** (1) A silent M-RoPE / inputs_embeds interaction that fabricated an 8σ false negative; (2) episode-stream train/validation leakage that is invisible at the episode level (2% of validation steps) but large at the shared-task-goal level (24%–85%); (3) pseudo-replicated significance testing whose anti-conservativeness we measure on synthetic clustered nulls (21.2% type-I error at nominal 5%).
- **A repaired, releasable protocol.** Pinned-revision, goal-disjoint frozen train/validation/test manifests with an untouched test region beyond every historical data access; episode-clustered paired inference with per-seed effects and validated calibration; and a prespecified confirmatory analysis committed before test access. All artifacts, per-example predictions, and analysis code are released.
- **A measured cost of getting evaluation wrong.** Because we ran the same ablation twice — once under the flawed protocol, once under the repaired one — we can report what the flaws actually bought: not inflated *p*-values alone, but a reversal of which mechanism wins, which action class carries the gain, whether our own hypothesis was refuted, whether the pipeline is deployable, and whether the control holds (Section 7.4).

2 Related Work

End-to-end GUI agents. The dominant paradigm treats action type, arguments, and coordinates as one autoregressive output. UI-TARS [13] exemplifies this with a unified action space across platforms and explicit thought traces. OS-Atlas [11] scales the same paradigm with 13M cross-platform GUI grounding instances and shows that action-label harmonization across web/Android/desktop is nontrivial. ShowUI [7] builds a compact VLA around Qwen2-VL with interleaved vision-language-action streaming. Our flat baseline (Variant A) replicates the joint-decoding paradigm of these systems.

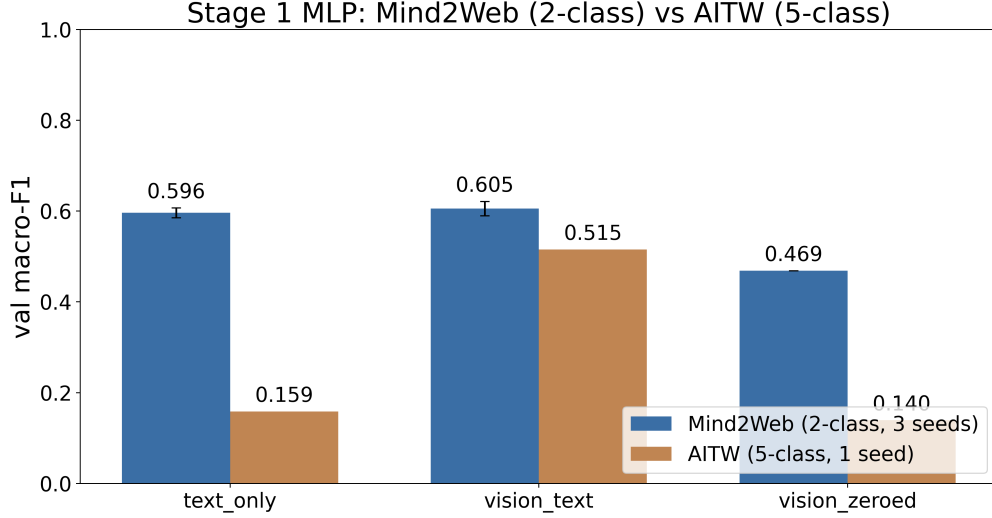


Figure 1: Stage 1 action-type classifier macro-F1 on Mind2Web (2 active classes) and AITW (5 active classes). Vision is essential on AITW (+0.414 over text-only) but redundant on Mind2Web (+0.009), because AITW instructions describe *goals* rather than physically specifying the action.

Modular grounding. SeeClick [4] argues that grounding is the central bottleneck in visual GUI agents. UGround / SeeAct-V [6] separates planning from grounding at the text-vs-pixel boundary. GUI-Actor [14] replaces coordinate-token regression with an attention-based action head driven by a dedicated <ACTOR> token plus a verifier. Our work studies a different module boundary: between *action-type prediction* and *spatial grounding*, not between planning text and coordinates.

Reward decomposition. UI-R1 [8] applies RL fine-tuning with a rule-based reward that separates action-type accuracy from action-argument accuracy on AndroidControl. It is the closest training analog: type and location as separable optimization targets; we use supervised conditioning rather than RL. RegionFocus [2] refines grounding at inference; we change the training-time conditioning pathway. SayCan [1] and ReAct [12] provide the modular-decision precedent, and RT-2 [3] the unified token-level action generation paradigm we critically examine.

Evaluation integrity. Our methodological findings echo, in a GUI setting, long-standing concerns about test-set contamination and dependence-blind inference: grouped data require grouped splits and cluster-aware uncertainty, and repeated measurements of the same example must not be pooled as independent evidence. Mind2Web’s own metric design (element accuracy vs. operation F1 vs. step success) [5] reflects the type/location decomposition we study.

3 Data and Splits

3.1 Sources and preprocessing

We use two datasets under a canonical 8-class action taxonomy (*click*, *double_click*, *type*, *scroll*, *drag*, *hotkey*, *wait*, *finished*).

AITW [9] (via the `cjfcst/AITW_General` HuggingFace mirror, revision pinned to 5c0dc713) is the primary Stage-2 benchmark: five active classes (*click* 55%, *scroll* 13%, *finished* 12%, *type* 11%, *hotkey* 9%). Screenshots in the mirror are raw RGB bytes without headers; we decode them against seven observed resolutions with a portrait-aspect fallback. DUAL_POINT events are split into taps (*click*) versus swipes (*scroll*) by normalized touch–lift distance (threshold 0.04). Two data mixes matter below: `all_with_coords` (tap, swipe, type – action type is spatially predictive because taps

target small elements while swipes are gestures) and `taps_and_swipes` (a control where knowing the type carries little information about *where*).

Mind2Web [5] (multimodal release) provides the Stage-1 cross-dataset comparison and a Stage-2 cross-dataset control; it is click/type-dominated with only two active classes at scale.

Coordinates. Targets are normalized to $[0, 1]^2$ and rendered as the string "`(x, y)`" on a $[0, 999]$ grid; training loss is masked to the coordinate tokens.

3.2 Two evaluation protocols

Exploratory protocol (Phases 1–7, historical). Our original runs streamed the AITW training split in mirror order, filtered by data mix, and took *consecutive slices*: the first n_{train} steps for training and the next n_{val} for validation. This is the kind of split that looks innocuous in code review and is not: episodes and, worse, repeated task goals straddle the boundary (Section 5.2). Every historical number in this paper is labeled exploratory and is re-analyzed with dependence-aware statistics.

Confirmatory protocol (Phase 9, frozen). We rebuilt the data pipeline around a frozen manifest: the first 12,000 `all_with_coords` steps of the pinned mirror revision, grouped by *normalized task goal* (episodes never span goals), split goal-disjointly into train (9,511 steps), validation (1,660), and an **untouched test region** (829): test groups are drawn exclusively from beyond stream index 5,500, past the furthest index any historical run ever loaded (5,250), so no test step was trained on, evaluated on, or inspected during the exploratory phase. Evaluation sets are frozen and identical for every variant and training size (250 validation / 600 test steps for `all_with_coords`; 200/400 for the control mix); training subsets are nested prefixes. The manifest, split hashes, 17 disclosed near-duplicate goal pairs (token Jaccard ≥ 0.8), duplicate checks, and the prespecified analysis (contrasts, metrics, corrections, failed-run rules) were committed to version control *before* any test-set evaluation ran. We claim “decisions frozen before test access,” documented by commit history – not third-party preregistration.

4 Methods

4.1 Stage 1: action-type classifier

A *frozen* Qwen2-VL-2B encodes screenshot+instruction; mean-pooled final hidden states (1536-d) feed a 3-layer MLP trained with class-weighted cross-entropy (AdamW, lr 10^{-4} , batch 128, 8 epochs, best epoch by validation macro-F1). Modes: `text_only` vs. `vision_text`.

4.2 Stage 2: five conditioning mechanisms

Stage 2 fine-tunes Qwen2-VL-2B with LoRA (rank 16 on q/k/v/o projections, $\approx 0.2\%$ of parameters) to emit the coordinate string; all variants share data, optimizer (lr 2×10^{-5} , 2 epochs, batch 1), and evaluation. The only difference is how action-type information enters:

A (flat baseline). No conditioning; replicates joint decoding.

B (auxiliary loss). A linear action-type head on the pooled representation adds cross-entropy to the grounding loss ($\lambda=1$); *no action information at inference*.

C (hard routing). Trained to emit “`{word} (x, y)`”; at inference the gold action word is forced as the decode prefix.

D-hook (additive embedding). A learned table $E \in \mathbb{R}^{8 \times d}$; the row for the current action type is added to *all* token embeddings via a forward hook, preserving the normal `input_ids` path (embedding norm at convergence ≈ 0.07 at the headline setting).

D-token (prepended token – the original hypothesis). A sentinel `<|action_slot|>` token is prepended to the instruction and its embedding replaced in place by $E[a]$, M-RoPE-correctly (the token is in `input_ids`); norm at convergence ≈ 2.2 , a strong routable signal.

historical cell	val steps	sharing a train episode	sharing a train goal
all_wc, $n=300$	250	4 (2%)	14 (6%)
all_wc, $n=500$	250	4 (2%)	47 (19%)
all_wc, $n=800$	250	5 (2%)	49 (20%)
all_wc, $n=1200$	250	4 (2%)	61 (24%)
all_wc, $n=2500$	250	10 (4%)	133 (53%)
all_wc, $n=5000$	250	2 (1%)	213 (85%)
taps_swipes, $n=1000$	200	0 (0%)	48 (24%)

Table 1: Leakage in the historical (exploratory) splits: fraction of validation steps sharing an episode or a normalized task goal with that cell’s training slice. Goal sharing, not episode straddling, is the dominant channel.

Objective. With screenshot s , instruction g , action type a , coordinate tokens $y_{1:T}$ and variant-specific conditioning c :

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, s, g, c), \quad (1)$$

with $c = \emptyset$ for A and B, $c = E[a]$ for D-hook/D-token, and the forced word for C; B adds $\lambda \text{CE}(h_{\phi}(\bar{z}), a)$ (Eq. 1 otherwise unchanged).

End-to-end pipeline. The deployable configuration trains D-hook, then a small MLP head on frozen-backbone features (Stage 1 in-job), and evaluates grounding under *predicted* types; gold-type results are oracle ceilings, never deployable claims.

4.3 Evaluation metrics

hit@ r : fraction of predictions within normalized L2 distance r of the target ($r \in \{0.05, 0.10, 0.25\}$; headline 0.10). **Mean L2:** mean normalized distance (parse failures scored at the $\sqrt{2}$ sentinel and counted as misses; parse rates reported). Statistical methodology is Section 5.3.

5 Three Evaluation Pitfalls, Quantified

5.1 Pitfall 1: M-RoPE vs. inputs_embeds

Qwen2-VL assigns 3D (temporal, height, width) position IDs to visual tokens via an `input_ids`-keyed path. Our first D-token implementation injected the action embedding by passing `inputs_embeds` directly, which silently bypasses that path: image-token positions were computed without the action slot in the sequence, misaligning multimodal position embeddings for *every* visual token. The observed effect was a confident 8σ “A beats D” result – a pure artifact. Rebuilding the same mechanism through the `input_ids` path (D-hook, D-token) erased the deficit. Any conditioning study on M-RoPE-family models that touches `inputs_embeds` should verify position-ID computation explicitly; the artifact is silent, large, and directionally misleading.

5.2 Pitfall 2: stream-slice splits leak task goals

Table 1 quantifies the historical protocol’s leakage by reconstructing, for every historical cell, which episodes and normalized task goals each validation step shared with that cell’s training slice. Episode-level overlap – the leakage one usually checks – is small (2% at the headline cell). The dominant channel is *shared task goals*: 24% of headline validation steps have a goal string that also appears in training, rising to 85% at $n_{\text{train}}=5000$ – AITW goals repeat across many episodes, so consecutive-slice splitting almost guarantees template overlap at scale. Val-set difficulty therefore varies with n_{train} (each size had a different val slice), quietly confounding the historical scaling curve. The confirmatory protocol removes both channels by construction (goal-disjoint splits; Section 3.2).

Variant	Mind2Web macro-F1	AITW macro-F1
text_only	0.597 $\pm$ 0.011	0.141 $\pm$ 0.032
vision_text	0.605 $\pm$ 0.016	0.555 $\pm$ 0.036
vision_zeroed	0.469 $\pm$ 0.000	0.110 $\pm$ 0.051
Vision delta	+0.009	+0.414
TF-IDF + LogReg	0.622	0.168
Majority floor	0.472	0.140

Table 2: Stage 1 action-type classification (3 seeds, mean $\pm$ std; exploratory splits). On Mind2Web the instruction text alone contains the action type; on AITW it does not, and vision contributes +0.414 macro-F1. `vision_zeroed` collapsing to the majority floor is the sanity check that features, not priors, drive the result.

182 5.3 Pitfall 3: pseudo-replicated significance

183 Our original headline significance pooled per-example distances across seeds – treating 3×250
184 (seed, example) rows as 750 independent paired observations. Predictions of the same example
185 by different seeds are strongly correlated, and examples within an episode share state; the pooled
186 bootstrap therefore overstates evidence. On synthetic nulls matching the data’s dependence structure
187 (episode and example effects shared across seeds; zero true difference), the pooled test rejects 21.2%
188 of the time at nominal 5% (1000 simulations), while the replacement below achieves 5.1%.

189 **Repaired inference.** For each contrast we (i) average each example’s paired difference over seeds,
190 (ii) cluster examples by episode, (iii) bootstrap clusters (10,000 resamples) for 95% CIs, and (iv) use
191 an episode-level sign-flip permutation test (10,000 permutations; p -values reported no smaller than
192 Monte Carlo resolution – never $p=0$). **Estimand.** This tests the *fixed trained checkpoints* on new
193 data from the eval population. It is *not* calibrated for claims about the training procedure: adding
194 run-level (seed) offsets to the synthetic null drives the episode-clustered test to 55% rejection (and
195 the pooled test to 70%). With only three seeds per variant we therefore report per-seed means every-
196 where and make no procedure-level significance claims. The retracted pooled p -values (including a
197 headline $p=0.0002$) should not be cited.

198 6 Exploratory Results (historical splits, repaired statistics)

199 All numbers in this section come from the leaky historical splits and are therefore *exploratory*;
200 they are shown with episode-clustered statistics, and Section 7 gives the prespecified untouched-test
201 versions of every primary claim.

202 6.1 Stage 1: vision is essential exactly where instructions under-specify the action

203 Table 2 and Figure 1: on Mind2Web, instructions name the action (“click the submit button”), so
204 vision adds +0.009 macro-F1 over text; on AITW, instructions state goals (“open a new Chrome
205 private window”), text-only collapses toward the majority class, and vision adds +0.414. A TF-IDF
206 ceiling of 0.168 on AITW confirms text is uninformative there regardless of classifier strength. This
207 asymmetry decides where Stage-2 conditioning could matter at all.

208 6.2 Main ablation under clustered statistics

209 Table 3 re-scores the historical headline with the repaired statistics (per-seed values in Figure 3).
210 Three observations. First, the auxiliary-loss effect survives: B–A on hit@0.10 is +0.051 [+0.020,
211 +0.081] ($p = 0.0043$), where the invalid pooled analysis had claimed $p = 0.0002$ – the conclusion
212 stands but the evidence was overstated by an order of magnitude. Second, D-hook weakens to
213 marginal after multiplicity correction. Third, the comparison our proposal cared about most – D-
214 token vs. B – is *no longer significant* (−0.036 [−0.077, +0.005], $p = 0.114$): on the historical data

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p
A (flat)	0.255 ± 0.021	0.392	—	—
B (aux loss)	0.305 ± 0.012	0.362	+0.051 [+0.020, +0.081]	0.0043
C (hard routing)	0.279 ± 0.032	0.412	+0.024 [-0.016, +0.063]	0.278
D-hook (additive)	0.300 ± 0.028	0.365	+0.045 [+0.011, +0.085]	0.0227
D-token (prepended)	0.269 ± 0.025	0.375	+0.015 [-0.018, +0.047]	0.437

Table 3: Exploratory main ablation (AITW `all_with_coords`, $n_{\text{train}}=1200$, 3 seeds, historical leaky splits). Episode-clustered Δ vs. A with 95% cluster-bootstrap CIs and sign-flip permutation p . Under the repaired analysis, B survives (Holm-adjusted $p = 0.017$ across the four vs.-A contrasts); D-hook is marginal after correction (Holm $p = 0.068$); C and D-token are not significant.

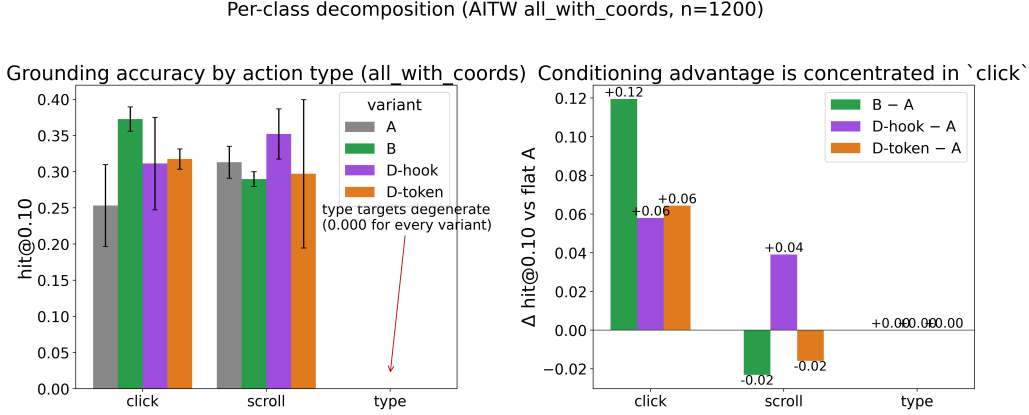


Figure 2: Exploratory hit@0.10 by action class (`all_with_coords`). The conditioning advantage concentrates in `click`; `type` is degenerate on AITW (all variants score exactly 0.000, mean L2 at the $\approx \sqrt{2}$ corner-to-corner distance).

we cannot even claim the prepended embedding is significantly *worse* than the auxiliary loss, only that it shows no sign of being better.

6.3 Per-class decomposition: the gain is click disambiguation

Breaking hit@0.10 down by class (Figure 2): B gains +0.120 on click (68% of examples) with a small scroll cost; C has the largest click gain (+0.122) but destroys scroll (−0.144) by forcing a decode prefix that derails the gesture class; D-hook alone improves both (click +0.058, scroll +0.039). The `type` class scores exactly 0.000 across all variants with mean normalized L2 at or slightly above $\sqrt{2}$, the corner-to-corner distance (a few predictions land outside the unit square): AITW `type` events carry degenerate touch coordinates and are ungroundable by any model. This corrects our earlier “`type`→text-field localization” interpretation: the real signal is **click-vs-scroll/type disambiguation** for the precise-target class, and the “`type`” rows merely add a constant drag that cancels in deltas.

6.4 Control conditions

On `taps_and_swipes` (action type spatially uninformative), the historical 3-seed means show no conditioned variant meaningfully above A (A 0.390 ± 0.008 , D-hook 0.392 ± 0.035 – a difference well inside one seed-std – with B 0.368 ± 0.009 and C 0.325 ± 0.032 at or below A on hit@0.10); per-example logs were not retained for A on this mix, so the clustered version of this control appears only in the confirmatory rerun (Table 6). On Multimodal-Mind2Web (click/type-dominated), D-hook vs. A was −0.006 hit@0.10 / +0.018 hit@0.25 over 2 seeds – a null, retained here deliberately: conditioning does not help where action type carries no spatial information, on either dataset.

contrast	Δ hit@0.10 [95% CI]	Δ hit@0.25 [95% CI]	perm. p (hit@0.10)
wrong – gold	−0.192 [−0.233, −0.149]	−0.329 [−0.372, −0.284]	$< 10^{-4}$
zero – gold	−0.093 [−0.128, −0.059]	−0.112 [−0.145, −0.079]	$< 10^{-4}$

Table 4: Exploratory D-token sensitivity (episode-clustered, 3 seeds): evaluating the same trained model with wrong or zeroed action embeddings severely degrades grounding. This establishes the embedding is *used* at inference – a sensitivity/oracle analysis, not evidence that conditioning caused any gain over A.

235 6.5 Low-data behavior: no clean curve

236 Our proposal predicted the advantage is a low-data prior. The historical grid ($n \in \{300, 500, 800\}$,
237 3 seeds) does *not* show a clean monotone: B–A on hit@0.10 is +0.067 at $n=300$ but +0.011 at
238 $n=500$ and −0.040 at $n=800$; D-hook is more stable (+0.069, +0.001, +0.048) and leads hit@0.25
239 and mean L2 at every low-data size. The two single-seed high-end cells disagree with each other:
240 at $n=5000$ the deltas shrink and partially reverse (B–A hit@0.25 goes negative), while at $n=2500$
241 D-hook posts the *largest* conditioning deltas anywhere in the study (+0.076 hit@0.10, +0.204
242 hit@0.25) – single-seed noise on a differently-leaky val slice, cutting in the direction *favorable*
243 to conditioning. We report all of this as: *inconclusive on the low-data-prior hypothesis – noisy,*
244 *non-monotone, and confounded by per-size val sets* (Figure 4); the prespecified low-data claim is
245 evaluated on the frozen test set in Section 7. An earlier draft of this work called the low-data
246 prediction “confirmed”; that claim did not survive this audit.

247 6.6 The e2e pipeline claim also weakens

248 The historical end-to-end result (predicted-type pipeline 0.271 vs. A 0.255 hit@0.10) was previ-
249 ously reported as a win. Under clustered statistics the paired difference is +0.016 [−0.017, +0.052]
250 ($p = 0.424$) – not significant; the oracle-type version is marginal (+0.037 [+0.005, +0.076]). The
251 deployable-pipeline question is therefore decided by the confirmatory run (P4, below), not the his-
252 torical data.

253 6.7 D-token causal-use sensitivity

254 Feeding the trained D-token model a *wrong* action id collapses grounding, and zeroing the embed-
255 ding hurts substantially (Table 4). The refutation of the proposal’s mechanism is therefore precise:
256 the prepended embedding is causally used, it is simply not the best way to exploit action-type super-
257 vision. Figure 5 shows one example where correct conditioning concentrates more attention mass
258 on the screenshot; across the three examples we probed the pattern is not uniform, so we present it
259 strictly as illustration, not evidence.

260 7 Confirmatory Results (untouched test, prespecified)

261 All variants were retrained from scratch on the frozen goal-disjoint splits (identical hyperparameters,
262 seeds 42–44) and evaluated once on the untouched 600-step test set; the four primary contrasts, the
263 headline metric (hit@0.10), the clustered test, and Holm correction were committed before any test
264 evaluation ran (Section 3.2).

265 **P1 (auxiliary supervision) — null.** B vs. A on hit@0.10: +0.011 [−0.005, +0.029], permutation
266 $p = 0.23$, Holm $p = 0.309$; mean L2 −0.005 [−0.016, +0.006] ($p = 0.42$). The variant the
267 exploratory analysis identified as the winner does *not* replicate. Auxiliary action-type supervision,
268 which adds no information at inference, leaves grounding statistically unchanged on data whose task
269 goals were never seen in training.

270 **P2 (inference-time conditioning) — positive, but oracle.** D-hook vs. A: +0.057 [+0.035, +0.079]
271 (Holm $p = 0.0004$; mean L2 −0.074 [−0.100, −0.050]), consistent in sign and size across all three

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p (Holm)
A (flat)	0.219 ± 0.018	0.450	—	—
B (aux loss)	0.230 ± 0.004	0.445	+0.011 [-0.005, +0.029]	0.23 (0.309)
C (hard routing)	0.219 ± 0.019	0.498	+0.000 [-0.022, +0.021]	1
D-hook (additive)	0.276 ± 0.003	0.376	+0.057 [+0.035, +0.079]	$< 10^{-4}$ (0.0004)
D-token (prepended)	0.265 ± 0.004	0.388	+0.046 [+0.027, +0.067]	0.0002

Table 5: **Confirmatory** main ablation on the untouched, goal-disjoint AITW test set ($n=600$ steps; $n_{\text{train}}=1200$; 3 seeds; mean $\pm$ std across seeds). Δ vs. A: episode-clustered 95% CI and permutation p (Holm-adjusted across the four primary contrasts where applicable).

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p
A (flat)	0.302 ± 0.007	0.268	—	—
B (aux loss)	0.316 ± 0.014	0.264	+0.013 [-0.013, +0.041]	0.364
C (hard routing)	0.236 ± 0.052	0.454	-0.067 [-0.100, -0.039]	$< 10^{-4}$
D-hook (additive)	0.349 ± 0.016	0.213	+0.047 [+0.010, +0.079]	0.0327

Table 6: Confirmatory control mix (`taps_and_swipes`) on its frozen 400-step test set. The control was designed as the falsifiable half of the thesis — conditioning should not help where action type is uninformative — and it *fails*: D-hook shows a significant gain here too (Section 7.2).

seeds. This is the one large effect that survives the clean split. It is an *oracle* result: D-hook receives the ground-truth action type at test time, which A never sees. P4 is what happens when that oracle is replaced by a real classifier.

P3 (the proposal’s mechanism) — reversed in our favour, and we flag it as such. D-token vs. B: +0.035 [+0.017, +0.054] (Holm $p = 0.0027$) — the prepended learned action embedding our original proposal advocated significantly *outperforms* the auxiliary loss on the clean test set. The exploratory data pointed the other way (Section 6), and the course-era version of this work reported the embedding as refuted. On the untouched test both embedding mechanisms beat B (D-token vs. A +0.046 [+0.027, +0.067], $p = 0.0002$; D-hook vs. B +0.046 [+0.026, +0.066], $p = < 10^{-4}$), while hard routing is exactly null (C vs. A +0.000 [-0.022, +0.021], $p = 1$). We stress that this contrast reverses in the direction that flatters our own hypothesis, which is precisely why it needed a prespecified test on data we had never touched.

P4 (deployable pipeline) — null. End-to-end predicted-type pipeline vs. A: +0.015 [-0.004, +0.034] (Holm $p = 0.309$). The reason is Stage 1 generalization, not Stage 2: the action-type classifier scores 0.795 accuracy on the frozen validation split but only 0.708 on the untouched test split, and the paired predicted-vs-oracle gap is -0.048 [-0.061, -0.035] ($p = < 10^{-4}$) — larger than the entire oracle advantage over A. **Classifier error consumes the whole benefit**, directly contradicting the historical claim that “classifier error is not the bottleneck.”

7.1 Where the confirmatory gain lives: scroll, not click

Table 8 overturns the exploratory mechanism story of Section 6.3. On leaky data the conditioning advantage looked like *click* disambiguation. On the untouched test set the dominant relative effect is

train size	A hit@0.10	Δ B–A [95% CI]	Δ D-hook–A [95% CI]
$n=300$	0.138	+0.038 [+0.013, +0.064]	+0.049 [+0.025, +0.073]
$n=500$	0.219	-0.004 [-0.027, +0.018]	+0.016 [-0.008, +0.039]
$n=800$	0.256	+0.003 [-0.014, +0.020]	+0.036 [+0.015, +0.057]

Table 7: Confirmatory low-data grid on the *same* frozen test set for every training size (nested training prefixes), removing the per-size-val-set confound of Figure 4.

variant	click ($n=393$)	scroll ($n=136$)	type ($n=71$)
A (flat)	0.312	0.064	0.000
B (aux loss)	0.337	0.042	0.000
C (hard routing)	0.310	0.113	0.000
D-hook (additive)	0.366	0.162	0.000
D-token (prepended)	0.367	0.108	0.000

Table 8: Confirmatory per-class hit@0.10 on the untouched test set (seed-averaged). The oracle conditioning gain is proportionally largest on *scroll* (A 0.064 $\rightarrow$ D-hook 0.162, a $2.5\times$ relative improvement) and modest on *click* (0.312 $\rightarrow$ 0.366). The auxiliary loss (B) improves click slightly while *harming* scroll. *type* remains exactly 0.000 for every variant: AITW type coordinates are degenerate.

on *scroll* (+0.098 absolute, $2.5\times$ relative), with a smaller click gain (+0.053). The mechanism this supports is different and simpler: swipe targets occupy a narrow, largely goal-independent region of the screen, so being told “this step is a scroll” relocates the prediction in a way the flat baseline cannot infer from the screenshot and instruction alone. B, which gets the same supervision only as a training signal, captures none of it and slightly harms scroll.

7.2 The control condition does not behave as designed

Our preregistered control assumed that on `taps_and_swipes` action type carries no spatial information, so conditioning should not help. It does help: D-hook vs. A on the control test set is +0.047 [+0.010, +0.079] ($p = 0.0327$), while B is null (+0.013 [-0.013, +0.041], $p = 0.364$) and C is significantly harmful (-0.067 [-0.100, -0.039], $p = <10^{-4}$). We report this as a **failed control**, not a second confirmation. It is coherent with Section 7.1: if the exploitable signal is “swipes land in a predictable region,” then a tap-versus-swipe mix is exactly a setting where action type *is* spatially informative, and our control was mis-designed rather than uninformative. The consequence for the paper is concrete: we cannot claim “conditioning helps if and only if action type is spatially predictive,” because we no longer have a matched condition in which it demonstrably is not.

7.3 Low data

The confirmatory low-data grid (Table 7) evaluates every training size against the *same* frozen test set, removing the per-size-val-set confound that made Figure 4 unreadable. D-hook’s oracle advantage is present at $n=300$ and $n=800$ and not resolved at $n=500$; B is positive only at $n=300$. There is no clean monotone low-data trend in either variant, so we again decline the “low-data prior” claim our proposal predicted.

Secondary contrasts, per-class decompositions, parse rates, and per-seed values for every run are in the released `results/phase9_rerun/rerun_analysis.json`. No procedure-level significance claim is made from three seeds (Section 5.3); per-seed means accompany every contrast.

7.4 What the broken evaluation actually cost

Comparing Section 6 with this section, the leaky split and the pooled bootstrap did not merely inflate significance. Every mechanistic conclusion we would have published flips:

- **Which variant wins.** Exploratory: auxiliary loss (B) beats the flat baseline and is the best mechanism. Confirmatory: B is indistinguishable from the baseline, and the two embedding mechanisms are the only ones that help.
- **Whether our own hypothesis was refuted.** Exploratory: D-token loses to B, so the proposal’s prepended embedding is refuted. Confirmatory: D-token significantly beats B.
- **Where the gain lives.** Exploratory: click disambiguation. Confirmatory: proportionally, scroll.

327 • **Whether the pipeline is deployable.** Exploratory: the predicted-type pipeline beats the
328 baseline and classifier error is not the bottleneck. Confirmatory: the pipeline is null and
329 classifier error consumes the entire oracle gain.

330 • **Whether the control holds.** Exploratory: conditioning is neutral where action type is un-
331 informative. Confirmatory: the control shows a significant effect, so the control is invalid.

332 A study can be internally consistent, multi-seed, ablated, controlled, and significant at $p < 0.002$,
333 and still have every one of its mechanistic claims reverse under a clean split and a calibrated test.

334 8 Limitations

335 The single positive confirmatory effect is oracle-gated: D-hook and D-token receive ground-truth
336 action types that a deployed agent would have to predict, and when we predict them the effect dis-
337 appears (P4). No claim in this paper should be read as a deployable improvement. Our control
338 condition failed (Section 7.2), so we have no valid matched setting in which action type is demon-
339 strably uninformative; the scope condition on when conditioning helps is therefore untested, and
340 designing a genuine control (for instance, targets randomized within class) is future work. The
341 scroll-versus-click mechanism reading in Section 7.1 is a post-hoc decomposition of confirmatory
342 data, not a prespecified test.

343 Offline step-level hit@radius is not long-horizon task success: it scores logged single steps, without
344 interaction, recovery, or compounding-error dynamics – the motivating failure mode is measured
345 here only via its step-level signature. The backbone is a single 2B model; conclusions are about
346 this scale and training recipe, not about GUI agents generally, and our baselines are small matched-
347 compute models trained by us, not state-of-the-art systems. The confirmatory estimand is fixed
348 checkpoints, not the training procedure (three seeds; per-seed effects reported). AITW’s type coor-
349 dinates are degenerate, so “grounding” on `all_with_coords` is effectively click/scroll grounding
350 plus a constant; the click-concentration analysis is post-hoc on exploratory data and only its prespec-
351 ified aggregate versions are confirmatory. Goal strings are our dependence proxy; 17 near-duplicate
352 train/test goal pairs (disclosed in the manifest) survive normalization, and subtler sharing (same app,
353 similar screens) is not modeled. The Mind2Web control used 2 seeds without per-example logs. Fi-
354 nally, the mirror we pin is a third-party repackaging of AITW; we verified revision stability but not
355 byte-level provenance against the original release.

356 9 Conclusion

357 On a test set whose task goals were never seen in training, the factorization thesis survives only
358 in oracle form. Telling a 2B GUI agent the ground-truth action type improves grounding (+0.057
359 [+0.035, +0.079] hit@0.10), and the prepended action embedding our proposal advocated is the
360 better of the two mechanisms that do this (D-token—B +0.035 [+0.017, +0.054]) — a reversal of the
361 course-era conclusion that the embedding was refuted. But auxiliary supervision alone is null, hard
362 routing is null, the effect concentrates in *scroll* rather than the click disambiguation we previously
363 reported, our control condition turns out not to be a control, and the only configuration one could
364 actually deploy — predicting the action type and conditioning on the prediction — is statistically
365 indistinguishable from doing nothing, because the classifier loses 0.795→0.708 accuracy on unseen
366 goals. The honest summary is that we measured a real but oracle-gated effect and no deployable
367 one.

368 The methodological result is the durable one. We ran the same five-variant ablation twice: once un-
369 der a protocol with three defects (an `inputs_embeds/M-RoPE` interaction fabricating an 8σ false
370 negative, goal-level split leakage reaching 85%, and a pooled bootstrap with 21.2% measured type-I
371 error), and once under a repaired one. The defects did not merely exaggerate significance; they
372 inverted which mechanism wins, which class carries the gain, whether our hypothesis was refuted,
373 whether the pipeline was deployable, and whether the control held. None of the three is exotic

— stream-sliced splits, embedding-injection shortcuts, and pooled per-seed bootstraps are ordinary practice — so we expect the same reversals are latent in other offline GUI-grounding evaluations. The frozen splits, per-example predictions, clustered-inference code, and prespecification are released so that both the object-level result and the audit can be checked and rerun.

Contributions

A. Chauhan: training and evaluation infrastructure, Stage-2 conditioning variants, cloud experiment orchestration, statistical reanalysis and confirmatory protocol, writing. **A. Ilyasov:** Stage-1 classifier study, data pipeline and error analysis, figures, editing. Both authors contributed to problem formulation and experimental design and approved the final manuscript.

Generative AI Usage Statement

Large-language-model assistants (Anthropic Claude) were used extensively under the authors’ direction: writing and debugging training, evaluation, and statistical-analysis code; orchestrating cloud experiments; drafting the integrity audit that led to Sections 5 and 7; and drafting and editing this manuscript. The authors specified the research questions and methodology, reviewed the analyses, verified reported numbers against the committed result artifacts (every table is generated programmatically from those artifacts), and take full responsibility for all content.

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Stage 2 ablation (AITW all_with_coords): $B \approx D\text{-hook} > C > A$

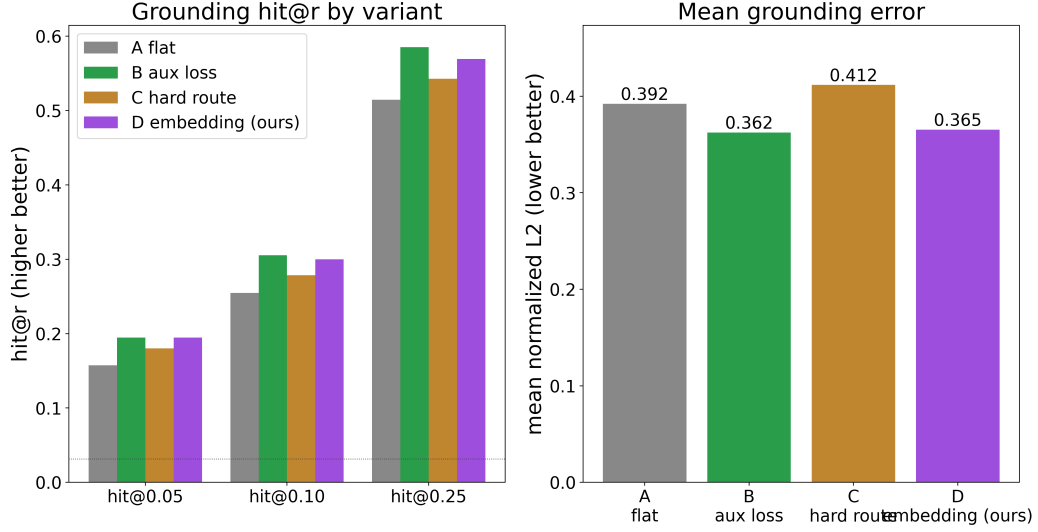


Figure 3: Exploratory ablation on `all_with_coords` (historical splits; per-seed values). Shown for continuity with the historical record; Table 3 carries the calibrated statistics and Table 5 the confirmatory test-set versions.

Data-scaling: action-type conditioning advantage vs training size
(AITW all_with_coords; 300-1200 are 3-seed means, 2500/5000 single seed)

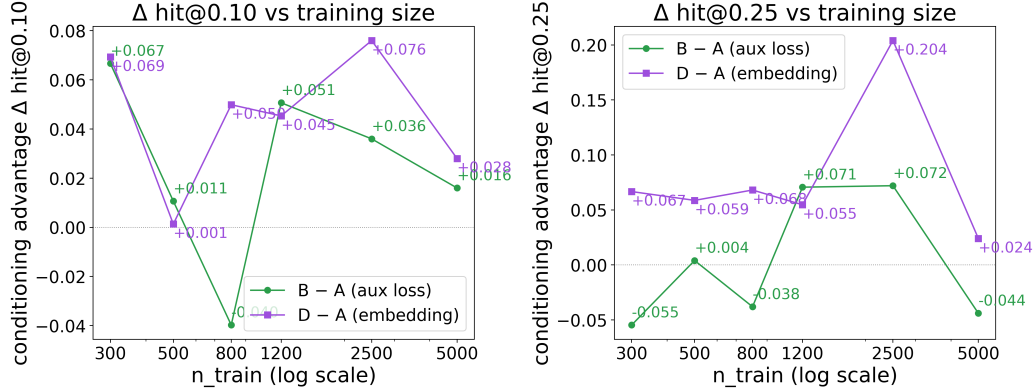


Figure 4: Exploratory conditioning advantage ($\Delta \text{hit@0.10}$ vs. A) across n_{train} . Caveat from Section 5.2: each historical size used a *different* val slice with different goal leakage (up to 85%), so this curve is directional at best; the confirmatory grid (Table 7) fixes one frozen test set for all sizes.

418 A Additional exploratory figures

419 These figures accompany Section 6 (exploratory results on the historical splits) and are referenced
420 from the main text.



Figure 5: Image-region attention for a held-out *click* example under three action-type conditionings (qualitative illustration; screenshot attention mass 0.204 under *click* vs. 0.132/0.110 under *type/scroll*).